

DIGITAL EVIDENCE ANALYSIS AND FILE RECOVERY USING SLEUTH KIT

EX. NO: 6

Date :

1. AIM

To conduct forensic analysis of a disk image using The Sleuth Kit (TSK), extract file-system metadata, identify and recover a deleted file, and generate a machine-readable forensic timeline body file.

2. SOFTWARE / TOOLS REQUIRED

Item	Details
Operating System	Windows
Forensic Tool	The Sleuth Kit (TSK) 4.14.0
Evidence Type	E01 forensic disk image
Evidence Files	4Dell Latitude CPi.E01 and 4Dell Latitude CPi.E02
File System	NTFS
Partition Offset	63 sectors
Deleted File	wizdata.dat
Metadata Address	10091

TSK Utilities Used

- fls – Lists files and directories, including deleted entries.
- mmls – Displays partition information.
- fsstat – Displays file-system statistics.
- istat – Displays metadata information for a specific metadata address.
- icat – Extracts file contents from a metadata address.
- mactime – Used to generate a human-readable timeline from a body file.

3. THEORY

The Sleuth Kit (TSK) is an open-source collection of command-line forensic tools used to examine disk images and file systems during digital investigations.

TSK allows investigators to analyse disk structures without modifying the original evidence. It provides tools for examining partitions, identifying file-system structures, listing allocated and deleted files, analyzing metadata, recovering file contents, and preparing information for forensic timeline analysis.

In this experiment, the disk image was analysed using the following workflow:

Disk Image → Partition Analysis → File-System Analysis → Deleted File Identification → Metadata Analysis → File Recovery → Body File Generation

The mmls utility identifies the partition layout and determines the sector offset required for subsequent analysis. The fsstat utility provides information about the file system. The fls utility is used to identify files and deleted entries. The istat utility provides metadata and timestamp information, while icat extracts data associated with a metadata address. The fls -m option can generate a machine-readable body file containing timestamp information for further timeline analysis.

4. PROCEDURE

Step 1: Directory Navigation and Version Verification

The Command Prompt was opened and the directory containing the TSK binaries was accessed. The installation was verified using the fls version command.

Command Executed

```
cd /d D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin  
fls -V
```

Output

The Sleuth Kit ver 4.14.0



```
C:\Windows\System32\cmd.exe  
D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>fls -V  
The Sleuth Kit ver 4.14.0  
D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>
```

Figure 1: Verification of The Sleuth Kit installation and version using the fls -V command.

Step 2: Partition Analysis Using mmls

The mmls command was used to analyze the partition structure of the forensic disk image.

Command Executed

```
mmls.exe "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01"
```

The output identified the primary NTFS partition beginning at sector 63.

```

C:\Windows\System32\cmd.e  x  +  v

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>fsstat.exe -o 63 "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" > filesystem_info.txt

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>type filesystem_info.txt
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)            Size: 0-256    Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256    Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident
$BITMAP (176)              Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)       Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)      Size: 8-8      Flags: Resident
$EA (224)                  Size: 0-65536   Flags:
$logGED_UTILITY_STREAM (256) Size: 0-65536   Flags: Non-resident

```

Figure 2: Execution of the mmls command identifying the NTFS partition starting at sector 63.

Step 3: File-System Statistics Using fsstat

The fsstat command was executed with the partition offset identified in the previous step.

Command Executed

```
fsstat.exe -o 63 "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" > filesystem_info.txt
```

The generated file-system information was viewed using:

```
type filesystem_info.txt
```

The output identified the file system as NTFS.

Important information obtained included:

- File System Type: NTFS
- Volume Serial Number: B26CB1CE6CB18D9B
- OEM Name: NTFS
- Version: Windows XP
- Sector Size: 512 bytes
- Cluster Size: 512 bytes

```
C:\Windows\System32\cmd.e X + v
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident, Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>fls.exe -o 63 -r "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" > file_list.txt

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>findstr /i "wizdata.dat" file_list.txt
++++ r/- * 0: wizdata.dat
++++ r/- * 10091: wizdata.dat
+++ r/r 10100-128-3: wizdata.dat
```

Figure 3: Generation of filesystem_info.txt and verification of the NTFS file-system statistics using the fsstat command.

Step 4: File-System Analysis and Deleted File Identification

The fls command was used recursively to list the files and directories contained in the partition.

Command Executed

fls.exe -o 63 -r "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" > file_list.txt

The generated file list was searched for wizdata.dat using: findstr /i "wizdata.dat" file_list.txt

```
C:\Windows\System32\cmd.e X + v
++ r/- * 0: ~DF992E.tmp
++ r/- * 0: ~DF9936.tmp
++ r/- * 0: ~DF99E3.tmp
++ r/- * 0: ~DF99EB.tmp
++ r/- * 346: ~DF99EB.tmp
+ r/- * 0: vb.ini
++ r/- * 0: smallfg.fon
+ r/- * 0: ~GLC8080.TMP
+ r/- * 0: ~GLH8080.TMP
+ r/- * 0: ~GLH8081.TMP

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>findstr /i "wizdata.dat" file_list.txt
++++ r/- * 0: wizdata.dat
++++ r/- * 10091: wizdata.dat
+++ r/r 10100-128-3: wizdata.dat

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>istat.exe -o 63 "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" 10091 > metadata_info.txt

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>type metadata_info.txt
MFT Entry Header Values:
Entry: 10091 Sequence: 4
$LogFile Sequence Number: 29681872
Allocated Directory
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags:
Owner ID: 0
Security ID: 310 (S-1-5-21-2009478354-688789844-1708537768-1003)
Created: 2004-08-20 20:48:07.255476800 (India Standard Time)
File Modified: 2004-08-20 20:48:09.628889600 (India Standard Time)
MFT Modified: 2004-08-20 20:48:09.628889600 (India Standard Time)
Accessed: 2004-08-20 20:48:09.628889600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Directory, Read Only
Name: ARCHIVE
Parent MFT Entry: 330 Sequence: 330
Allocated Size: 0 Actual Size: 0
Created: 2004-08-20 20:48:07.255476800 (India Standard Time)
File Modified: 2004-08-20 20:48:07.255476800 (India Standard Time)
MFT Modified: 2004-08-20 20:48:07.255476800 (India Standard Time)
Accessed: 2004-08-20 20:48:07.255476800 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 80
Type: $INDEX_ROOT (144-1) Name: $I30 Resident size: 440

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>
```

Figure 4: Identification of the deleted wizdata.dat file at NTFS metadata address 10091 using the fls command.

Step 5: File Metadata Analysis Using istat

The istat command was used to examine NTFS metadata address 10091.

Command Executed

```
istat.exe -o 63 "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" 10091 > metadata_info.txt
```

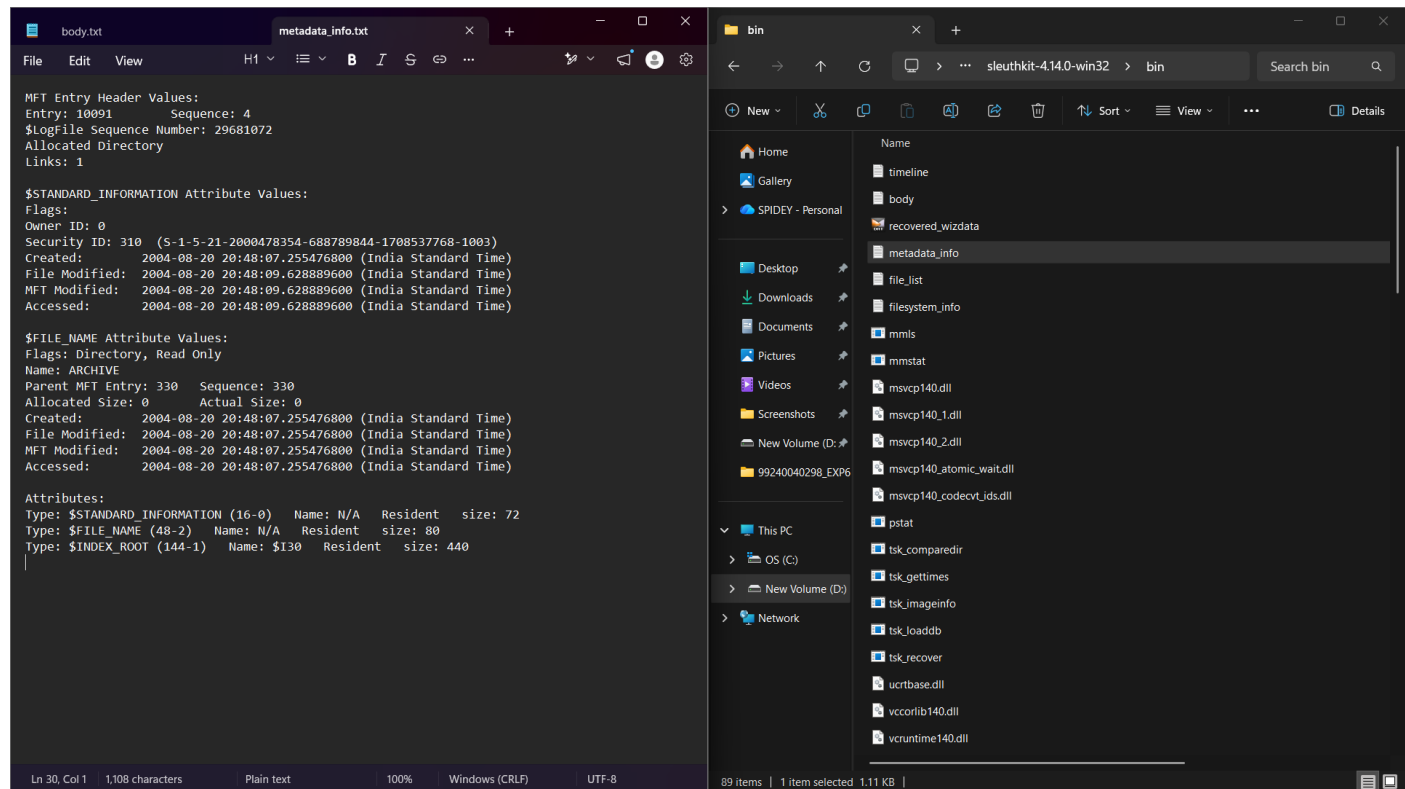


Figure 5: Extraction of MFT metadata and MAC timestamps for NTFS metadata address 10091 using the istat command.

Step 6: Recovery of the Deleted File Using icat

The icat command was used to extract data associated with metadata address 10091 from the forensic image.

Command Executed

```
icat.exe -o 63 "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" 10091 > recovered_wizdata.dat
```

The recovered file was verified using: `dir recovered_wizdata.dat`

The output confirmed that the recovered file was created successfully with a size of 440 bytes.

```
C:\Windows\System32\cmd.exe X + v

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 80
Type: $INDEX_ROOT (144-1) Name: $I30 Resident size: 440

D:\nira\99240040298_EXP6\slleuthkit-4.14.0-win32\bin>type metadata_info.txt
MFT Entry Header Values:
Entry: 10091 Sequence: 4
$LogFile Sequence Number: 29681072
Allocated Directory
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags:
Owner ID: 0
Security ID: 310 (S-1-5-21-2000478354-688789844-1708537768-1003)
Created: 2004-08-20 20:48:07.255476800 (India Standard Time)
File Modified: 2004-08-20 20:48:09.628889600 (India Standard Time)
MFT Modified: 2004-08-20 20:48:09.628889600 (India Standard Time)
Accessed: 2004-08-20 20:48:09.628889600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Directory, Read Only
Name: ARCHIVE
Parent MFT Entry: 330 Sequence: 330
Allocated Size: 0 Actual Size: 0
Created: 2004-08-20 20:48:07.255476800 (India Standard Time)
File Modified: 2004-08-20 20:48:07.255476800 (India Standard Time)
MFT Modified: 2004-08-20 20:48:07.255476800 (India Standard Time)
Accessed: 2004-08-20 20:48:07.255476800 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 80
Type: $INDEX_ROOT (144-1) Name: $I30 Resident size: 440

D:\nira\99240040298_EXP6\slleuthkit-4.14.0-win32\bin>icat.exe -o 63 "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" 10091 > recovered_wizdata.dat

D:\nira\99240040298_EXP6\slleuthkit-4.14.0-win32\bin>dir recovered_wizdata.dat
Volume in drive D is New Volume
Volume Serial Number is D05D-A148

Directory of D:\nira\99240040298_EXP6\slleuthkit-4.14.0-win32\bin

24-09-2026 00:52 440 recovered_wizdata.dat
1 File(s) 440 bytes
0 Dir(s) 82,468,687,872 bytes free
```

Figure 6: Successful extraction and recovery of data associated with metadata address 10091 using the icat command.

Step 7: Generation of the Forensic Body File

A machine-readable body file containing file-system timestamp information was generated using the fls command with the -m and -r options.

Command Executed

```
fls.exe -o 63 -m / -r "D:\nira\99240040298_EXP6\4Dell Latitude CPi.E01" > body.txt
```

The generated body file was verified using: dir body.txt

The file was approximately 3,249,309 bytes in size.

The contents were inspected using: more body.txt

The body file contained machine-readable file-system records and MAC timestamp information for forensic timeline analysis.

```
C:\Windows\System32\cmd.exe + v
D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>fls.exe -o 63 -m / -r "D:\nira\99240040298_EXP6\4Dell Latitude Cpi.E01" > body.txt
D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>dir body.txt
Volume in drive E is New Volume
Volume Serial Number is D85D-A148

Directory of D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin

24-09-2026  00:52                3,249,389 body.txt
               1 File(s)                3,249,389 bytes
               0 Dir(s)                82,465,435,648 bytes free

D:\nira\99240040298_EXP6\sleuthkit-4.14.0-win32\bin>more body.txt
0|/Documents and Settings|($FILE_NAME)|3671-48-2|d/drwxrwxrwx|0|0|118|1092934741|1092934741|1092934741|1092934741
0|/Documents and Settings|3671-144-7|d/drwxrwxrwx|0|0|56|1093619285|1092956645|1092956645|1092934741
0|/Documents and Settings|All Users|($FILE_NAME)|3673-48-2|d/drwxrwxrwx|0|0|84|1092934741|1092934741|1092934741|1092934741
0|/Documents and Settings|All Users|3673-144-7|d/drwxrwxrwx|0|0|56|1093619286|1092954817|1092954817|1092934741
0|/Documents and Settings|All Users|Application Data|($FILE_NAME)|3787-48-2|d/d-x-x-x-x|0|0|98|1092934769|1092934769|1092934769|1092934769
0|/Documents and Settings|All Users|Application Data|3787-144-7|d/d-x-x-x-x|0|0|56|1093619307|1092934809|1092934809|1092934769
0|/Documents and Settings|All Users|Application Data|desktop.ini|($FILE_NAME)|3808-48-2|r/r-r-xr-xr-x|0|0|88|1092934809|1092934809|1092934809|1092934809
0|/Documents and Settings|All Users|Application Data|desktop.ini|3808-128-1|r/r-r-xr-xr-x|0|0|62|1093620583|1092934809|1092934809|1092934809
0|/Documents and Settings|All Users|Application Data|Microsoft|($FILE_NAME)|3708-48-2|d/drwxrwxrwx|0|0|84|1092934769|1092934769|1092934769|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|3708-144-7|d/drwxrwxrwx|0|0|56|1093621576|1093015544|1093015544|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|($FILE_NAME)|3709-48-2|d/drwxrwxrwx|0|0|78|1092934769|1092934769|1092934769|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|3709-144-1|d/drwxrwxrwx|0|0|224|1093015078|1092934770|1092934770|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|DSS|($FILE_NAME)|3712-48-2|d/drwxrwxrwx|0|0|72|1092934770|1092934770|1092934770|1092934770
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|DSS|3712-144-1|d/drwxrwxrwx|0|0|256|1093015078|1092934770|1092934770|1092934770
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|DSS|MachineKeys|($FILE_NAME)|3713-48-2|d/drwxrwxrwx|0|0|88|1092934770|1092934770|1092934770|1092934770
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|DSS|MachineKeys|3713-144-1|d/drwxrwxrwx|0|0|48|1093015078|1092934770|1092934770|1092934770
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|RSA|($FILE_NAME)|3710-48-2|d/drwxrwxrwx|0|0|72|1092934769|1092934769|1092934769|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|RSA|3710-144-1|d/drwxrwxrwx|0|0|256|1093015078|1092934769|1092934769|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|RSA|MachineKeys|($FILE_NAME)|3711-48-2|d/drwxrwxrwx|0|0|88|1092934769|1092934769|1092934769|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|Crypto|RSA|MachineKeys|3711-144-1|d/drwxrwxrwx|0|0|48|1093015078|1092934769|1092934769|1092934769
0|/Documents and Settings|All Users|Application Data|Microsoft|Dr Watson|Dr Watson|($FILE_NAME)|10977-48-2|d/drwxrwxrwx|0|0|84|1093015544|1093015544|1093015544|1093015544
0|/Documents and Settings|All Users|Application Data|Microsoft|Dr Watson|10977-144-1|d/drwxrwxrwx|0|0|264|1093015548|1093015548|1093015548|1093015544
0|/Documents and Settings|All Users|Application Data|Microsoft|Dr Watson|Dr Watson|user.dmp|($FILE_NAME)|10980-48-2|r/rwxrwxrwx|0|0|82|1093015548|1093015548|1093015548|1093015548
0|/Documents and Settings|All Users|Application Data|Microsoft|Dr Watson|Dr Watson|drwtsn32.log|($FILE_NAME)|10979-48-2|r/rwxrwxrwx|0|0|90|1093015545|1093015545|1093015545|1093015545
0|/Documents and Settings|All Users|Application Data|Microsoft|Dr Watson|drwtsn32.log|10979-128-3|r/rwxrwxrwx|0|0|37044|1093015548|1093015548|1093015548|1093015545
0|/Documents and Settings|All Users|Application Data|Microsoft|Dr Watson|user.dmp|($FILE_NAME)|10980-48-2|r/rwxrwxrwx|0|0|82|1093015548|1093015548|1093015548|1093015548
0|/Documents and Settings|All Users|Application Data|Microsoft|Dr Watson|user.dmp|10980-128-3|r/rwxrwxrwx|0|0|28023|1093015548|1093015548|1093015548|1093015548
0|/Documents and Settings|All Users|Application Data|Microsoft|HTML Help|($FILE_NAME)|7325-48-2|d/drwxrwxrwx|0|0|94|1092955220|1092955220|1092955220|1092955220
0|/Documents and Settings|All Users|Application Data|Microsoft|HTML Help|7325-144-1|d/drwxrwxrwx|0|0|48|1093015078|1092955220|1092955220|1092955220
0|/Documents and Settings|All Users|Application Data|Microsoft|Media Index|($FILE_NAME)|7287-48-2|d/drwxrwxrwx|0|0|88|1092955087|1092955087|1092955087|1092955087
0|/Documents and Settings|All Users|Application Data|Microsoft|Media Index|7287-144-7|d/drwxrwxrwx|0|0|56|1093015078|1092956674|1092956674|1092955087
0|/Documents and Settings|All Users|Application Data|Microsoft|Media Index|wmplibrary_v_0_12.db|($FILE_NAME)|6074-48-2|r/rwxrwxrwx|0|0|186|1092955097|1092955097|1092955097|1092955097
0|/Documents and Settings|All Users|Application Data|Microsoft|Media Index|wmplibrary_v_0_12.db|6074-128-5|r/rwxrwxrwx|0|0|245760|1092956674|1092956674|1092955097
0|/Documents and Settings|All Users|Application Data|Microsoft|Network|($FILE_NAME)|4759-48-2|d/drwxrwxrwx|0|0|88|1092954089|1092954089|1092954089|1092954089
0|/Documents and Settings|All Users|Application Data|Microsoft|Network|4759-144-1|d/drwxrwxrwx|0|0|256|1093015078|1092954089|1092954089|1092954089
0|/Documents and Settings|All Users|Application Data|Microsoft|Network|Connections|($FILE_NAME)|4762-48-2|d/drwxrwxrwx|0|0|88|1092954089|1092954089|1092954089|1092954089
0|/Documents and Settings|All Users|Application Data|Microsoft|Network|Connections|4762-144-1|d/drwxrwxrwx|0|0|136|1093015078|1092954089|1092954089|1092954089
```

Figure 7: Generation and verification of the body.txt file containing machine-readable MAC timestamp records.

5. OBSERVATIONS / FORENSIC FINDINGS

1. The Sleuth Kit version 4.14.0 was successfully verified.
2. The mmfs command successfully identified the NTFS partition beginning at sector 63.
3. The fsstat command confirmed that the file system was NTFS.
4. Recursive fls analysis identified the deleted file wizdata.dat and associated it with metadata address 10091.
5. The istat command successfully extracted MFT metadata and MAC timestamps for metadata address 10091.
6. The icat command successfully extracted data associated with metadata address 10091 and created recovered_wizdata.dat.
7. The recovered file was verified to have a size of 440 bytes.
8. The fls -m command successfully generated body.txt, containing machine-readable file-system and timestamp records.
9. The mactime utility was unavailable in the configured Windows environment, so a formatted timeline.txt file was generated.

6. LIMITATION

The mactime utility was not available in the configured Windows environment. The command:

```
mactime.exe -h
```

returned an error indicating that the command was not recognized.

As a result, the machine-readable body.txt file was successfully generated, but the final formatted human-readable timeline using mactime could not be produced.

7. CONCLUSION

The experiment successfully demonstrated the use of The Sleuth Kit (TSK) 4.14.0 for forensic disk-image analysis and deleted-file recovery. The partition structure and NTFS file system were identified, the deleted wizdata.dat entry was located, its associated metadata was examined, and the data was successfully recovered. A machine-readable body file was also generated to support forensic timeline analysis.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

8.RESULT

The 4Dell Latitude CPi.E01 forensic disk image was successfully analyzed using TSK. The NTFS partition was identified at sector 63, the deleted wizdata.dat file was identified at metadata address 10091, and its associated data was successfully extracted as recovered_wizdata.dat with a size of 440 bytes. The body.txt file containing MAC timestamp records was also generated successfully.